

be avoided. It improves the overall performance of the network environment.

In simple terms, the cloud is at the top—in the sky. The end user is at the bottom (on earth). Now, with fog computing, the computing infrastructure is near the end user (on earth). And since it is near the end system where the actual services are required, one does not need to access everything through the Internet.

The concepts of edge computing and mist computing are closely related to each other and are often used synonymously. Mist computing can be used as an intermediate layer between the cloud and the fog or between the cloud and the edge.

In the case of edge computing, devices with programmable automation controllers (PAC) are used, while mist computing uses lightweight computing objects with microchips and microcontrollers. The deployment of edge computing or mist computing is done at the location of the end user. Fog computing is the extension of cloud services to the edge of a network. Fog computing works for the delivery of cloud services in a simple and fast way, while edge computing can work without the fog or cloud. Mist computing is not a mandatory layer of fog computing and can work separately.

In a fog, we generally cannot see beyond 100 metres, but we do have some visibility within that range. In fog computing, the same concept applies. There is a range within which we have access to resources.

With fog computing, performance on the following parameters is much better:

- Latency
- Capacity
- Bandwidth
- Responsiveness
- Security
- Speed
- Robustness
- Fault tolerance
- Data integration
- Energy

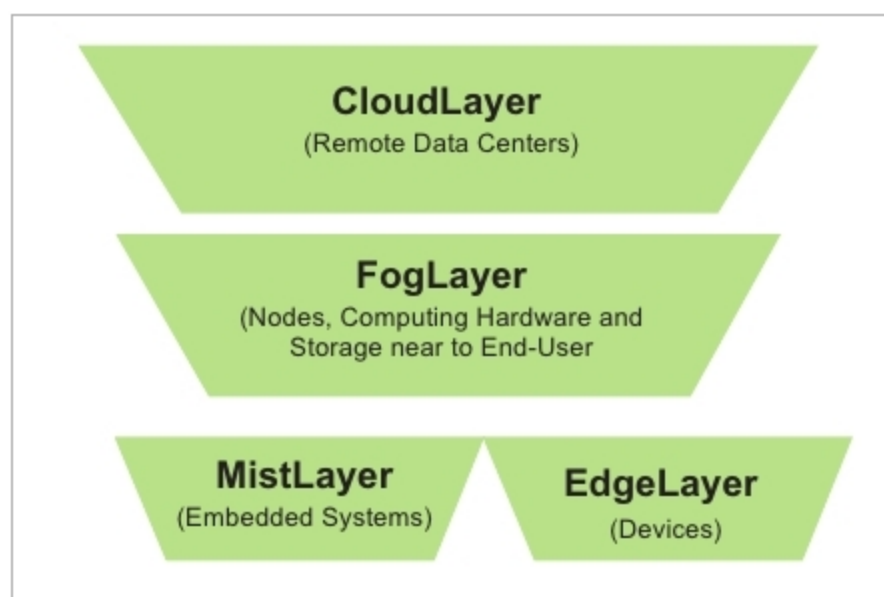


Figure 1: Layers of the cloud—the fog, mist and edge computing

Prominent libraries and tools for simulation of advanced networks

Table 1 lists the tools and libraries used for simulation of advanced networks. These include IoT, fog computing and edge computing.

Table 1

Title	URL
iFogSim	github.com/Cloudslab/iFogSim
EmuFog	github.com/emufog/emufog
FogTorch	github.com/di-unipi-socc/FogTorch
Clonix	clonix.net
IMUNES	imunes.net
Cooja	contiki-os.org
CupCarbon	cupcarbon.com
DSA	iot-dsa.org
Mininet	mininet.org
Node-RED	nodered.org
OpenIoT	openiot.eu
Unetlab	routereflector.com/unetlab
Zetta	zettajs.org
Shadow	shadow.github.io
GNS3	gns3.com/
IoTivity	iotivity.org
KAA	kaaproject.org
CORE	nrl.navy.mil/itd/ncs/products/core
NS3	nsnam.org
Netkit	netkit.org

iFogSim: This is a high-performance open source toolkit for fog computing, edge computing and IoT, which is used to model and simulate the networks of edge computing, the Internet of Things and fog computing. iFogSim integrates the resource management techniques that can be further customised as per the research area.

iFogSim works in association with CloudSim, a widely used library for the simulation of cloud based environments and for resource management. The CloudSim layer handles the events between the components of fog computing using iFogSim.

The following are the classes of iFogSim that are required to simulate the fog network:

- Fog device
- Sensor
- Actuator
- Tuple
- Application
- Monitoring edge
- Resource management service